**DIFF. B/W ANONYMOUS & SUB BLOCK :**

| ANONYMOUS BLOCK | SUB BLOCK |
|---|---|
| 1. UNNAMED BLOCK | 1. NAMED BLOCK |
| 2. THIS BLOCK CODE IS NOT SAVED IN DB. | 2. THIS BLOCK CODE IS SAVED IN DB AUTOMATICALLY. |
| 3. IT CANNOT REUSABLE. | 3. IT CAN BE REUSABLE. |
| 4. EVERY TIME COMPILATION OF CODE. | 4. PRE - COMPILED CODE ( FIRST TIME COMPILATION ONLY) |

**STRUCTURE OF PL/SQL BLOCK:**

```
DECLARE

BEGIN
    < WRITING SQL STATEMENTS>;
    < PL/SQL LOGICAL CODE>;
EXCEPTION
    < HANDLING EXCEPTIONS>;
END;
/
```

Exe

```
Declare
  X varchar2 10;
Begin
  X :=10
 Dbms_output.line (x)
End ;
```

&takes a input in runtime.

```
DECLARE
    x number(10);
    y number(10);
    z number(10);

BEGIN
    x := &x;
    y := &y;
    z := x+y;
    DBMS_OUTPUT.PUT_LINE(z);
END;
/
```

Select a,b in to x,y from emp where emp=1;

```
SET SERVEROUTPUT ON;
DECLARE
   eemp number(10);
   efirst_name varchar2(10);
BEGIN
    SELECT employee_id, first_name INTO eemp, efirst_name FROM HR.employees WHERE
ROWNUM = 1;
    DBMS_OUTPUT.PUT_LINE('eemp  '|| eemp || 'efirst  ' || efirst_name);
END;
/

    SELECT employee_id, first_name INTO eemp, efirst_name FROM HR.employees WHERE
EMPLOYEE_ID = &EMPLOYEE_ID;
```

– if database table datatype change from varchar to date, plsql program associate variable i have to change. Otherwise mismatch works.

```
Enter value for empno: 1122
DECLARE
*
ERROR at line 1:
ORA-01403: no data found
ORA-06512: at line 5


SQL> DECLARE
  2  v_ENAME VARCHAR2(10);
  3  v_SAL NUMBER(10);
  4  BEGIN
  5  SELECT ENAME,SAL INTO v_ENAME,v_SAL FROM EMP WHERE EMPNO=&EMPNO;
  6  DBMS_OUTPUT.PUT_LINE(v_ENAME||','||v_SAL);
  7  END;
  8  /
Enter value for empno: 7788
SCOTT,3000

PL/SQL procedure successfully completed.

SQL> DESC EMP;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 EMPNO                                     NOT NULL NUMBER(4)
 ENAME                                              VARCHAR2(10)
 JOB                                                VARCHAR2(9)
 MGR                                                NUMBER(4)
 HIREDATE                                           DATE
 SAL                                                NUMBER(7,2)
 COMM                                               NUMBER(7,2)
 DEPTNO                                             NUMBER(2)

SQL>
```

VA

DATE

DATE

Automatically v ename datatype changes , advantages **of anchor notations:**

**Column level**

anchor notations

```
PRG:
====
DECLARE
v_ENAME EMP.ENAME%TYPE;
v_SAL EMP.SAL%TYPE;
BEGIN
SELECT ENAME,SAL INTO v_ENAME,v_SAL FROM EMP WHERE EMPNO=&EMPNO;
DBMS_OUTPUT.PUT_LINE(v_ENAME||','||v_SAL);
END;
/
OUTPUT:
Enter value for empno: 7788
SCOTT,3000
```

Rowlevel attributes

```
EX:         i     EMP%ROWTYPE;

PRG:
DECLARE
i EMP%ROWTYPE;
BEGIN
SELECT ENAME,SAL INTO i.ENAME,i.SAL FROM EMP WHERE EMPNO=&EMPNO;
DBMS_OUTPUT.PUT_LINE(i.ENAME||','||i.SAL);
END;
/
```